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## Element H

### Testing Plan

#### **High Priority:**

##### Safety -

To determine how safe the lights are, we can calculate the percentage of lights that are out of reach of children. We can ensure that no kids are tall enough to reach a majority of them. If more than 100% of lights *are* out of reach, that would score a 4. If 75%+ of lights are out of reach, that would score a 3. If 50%+ of lights are out of reach, that would score a 3. If less than 50% are out of reach, that would be a 1.

##### Aesthetic/Creates a diverse mood -

Although this is a bit of an unconventional category to “test,” because it is one of our high priority requirements, we want a way to measure how much our lights meet the goal. We can easily survey the adults in the room and ask them how well they think our lights perform and how good they look. We can create a survey for them to fill out, which they can then use to rate us on a scale of 1 to 4 on this criteria. The average score that we receive will be the overall score that represents our performance.

##### Durability -

Because it is so difficult to find a good time to go in and complete installation in the classroom, it is crucial to ensure that the lights stay functional for a long period of time. We will not be able to go in periodically to do maintenance on any lights that aren't working properly. This is simply not feasible and needs to be avoided at all costs. To measure how durable our lights are, we will approach it from a perspective of how long it takes the lights to experience any difficulties. If the lights last 5 years or longer, that achieves a score of 4. If the lights last 3-5 years, that will be a score of 3. If the lights last 1-3 years, that is a score of 2. Finally, if the lights last 1 year or less, this is a score of 1.

### Entertaining and Effectiveness -

Similarly to aesthetics, entertainment is a hard category to measure. What will happen is we will go to their classroom and show the kids and the stakeholders what the lights look like and the options of different colors and patterns. From there, we will note how the various people react, whether it's in a negative or positive way. This will help indicate and anticipate how kids will respond to the lights during instructional time.

### Adaptability to different needs -

In order to test this, we can communicate with our primary stakeholders and make sure that the lights have all of the features that they want them to have. We will also meet with the stakeholders to show them all the features that we have programmed into the lights and confirm that they have everything they want. We can also encourage the teachers at Whitemarsh to communicate with us how useful they find the lights to be and how often they are able to use them in order to satisfy multiple different functions.

## **Low Priority:**

### Ease of Use -

By observing how long it takes for teachers to complete different tasks and how easily they can navigate the app, we will get a better understanding of our achievement under this criteria. To quantify our score, we can give our stakeholders different tasks to complete using the app and time how long it takes them to do it. We can repeat this 5 times and take the average of our findings. Lower numbers will indicate better performance.

### Number of Zones -

The number of zones is a fairly simple criteria for us to measure. We can award our group 1 point per zone that we are able to provide. The more points we have, the better job we did at giving the teachers zoning capabilities.

### Affordability -

Our original target was to stay under \$100 for the total costs of the project. In order to determine how well we stayed within budget, we can award different scores depending on certain criteria. We can give ourselves: a 4 if we spent less than \$100, a 3 if we spent \$100, a 2 if we went up to 50% over budget, and a 1 if we went more than 50% over budget.

## Feedback on Procedures

### **Expert #1:** Mr. Muscarella

Mr. Muscarella is an experienced engineer and educator, so he has a unique ability to relate both to our overall project and the instructors at Whitemarsh. He also understands the importance of prioritizing both the students' and the teachers' needs. For these reasons, we had him look over our testing plans and give us feedback about how we can improve. He gave us important feedback about how we could increase the focus on the teachers and their ability to use and benefit from the lights. He reminded us of our overall goal across all criteria: to deliver a functional and useful product that is easy to use. This helped us better understand how to focus each criteria's testing around that main goal. For example, the number of zones we provide is important, but the bigger reason *why* it is important is because we want to make the lights useful and able to be used to satisfy varying needs.

### **Expert #2:** Ms. Upadhya

The second person we approached for feedback was Ms. Upadhya. Ms. Upadhya has worked diligently, completing a total of three courses in engineering and design. She has also taken part in national technology competitions and has received awards for her brilliant innovation. This fall, she plans to major in Aerospace Engineering at a national university and has been admitted to numerous high-rigor undergraduate engineering programs across the country. We were more than happy to receive her input on our project. After reviewing our testing procedures, she was able to provide extremely useful feedback that we will be taking into consideration while evaluating our project. She liked our overall ideas and concepts, as they would allow us to objectively score our final product and understand how well we achieved our major goals. However, she reminded us of the importance of taking bias into account while asking for feedback from teachers. Specifically, she recognized that our presence in

the room may lead to inaccurate evaluations from the teachers, as they may be afraid to be truthful about their opinions. This was something we hadn't properly considered before and helped us realize that we would have to be very careful when asking for feedback. In order to ensure the most accurate and effective responses, we will have to remind them to be honest and urge them to give us their true reactions to our final product. We are very grateful to have received this feedback, as without it, our results would not have been as reliable.